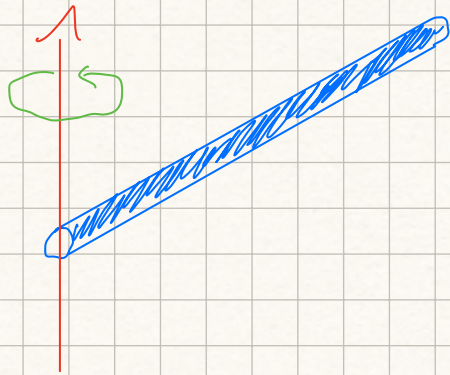


Moment of Inertia

Thin rod of length L and mass m , perpendicular to the axis of rotation, rotating about one end:



$$I = \frac{1}{3} mL^2$$

Parallel Axis Theorem

$$I = I_{cm} + md^2$$

Use when an object is not rotating around its CoM

Use composite modeling.

Sum the inertias of all the components to get the most accurate value.

For each component:

$$I_i = I_{cmi} + m_i d_i^2$$

where

I_{cmi} is the moment of inertia about its own center

d_i is the distance from the pivot point to the CoM

Total inertia:

$$I_{tot} = \sum_i I_i$$

Real inertia:

$$I_{rod} = \frac{1}{3} \cdot 0.68 \cdot (0.135)^2 = 0.004$$

$$I_{pb} = 0 + 0.15 \cdot (0.04)^2 = 0.00024$$

$$I_{md} = 0 + 0.03 \cdot (0.09)^2 = 0.000243$$

$$I_{stem} = 0 + 0.04 \cdot (0.09)^2 = 0.000324$$

$$I_{bb} = 0 + 0.1 \cdot (0.14)^2 = 0.00196$$

$$I_{tot} = 0.006767$$